

Basic Laws in Economics

Law of Diminishing Marginal Utility, Demand, Supply, and Market Equilibrium

Law of Diminishing Marginal Utility and Introduction to Demand

1. Meaning of Utility

In economics, utility means the want-satisfying power of a good or service. It is the satisfaction or benefit a consumer receives from consuming a good or service.

Examples:

- A glass of water gives utility to a thirsty person.
- A laptop gives utility to a CST student by helping with programming, assignments, communication, and online learning.
- Mobile data gives utility because it allows internet access.
- An online programming course gives utility by improving skills.
- AI tools give utility by helping with coding, writing, summarizing, and learning.

Utility is subjective. The same good may give different levels of satisfaction to different people.

For example, a high-performance laptop may give very high utility to a CST student, but lower utility to someone who only uses a computer for basic browsing.

Note

Why does the first glass of water feel more satisfying than the fourth glass? Why does the first hour of a programming tutorial often help more than the fifth continuous hour?

2. Total Utility and Marginal Utility

Total Utility

Total utility is the total satisfaction obtained from consuming all units of a good or service.

Example: If a student watches three tutorial videos and gets satisfaction from all three together, that combined satisfaction is total utility.

Marginal Utility

Marginal utility is the additional satisfaction obtained from consuming one more unit of a good.

Formula: $\text{Marginal Utility} = \text{Change in Total Utility} / \text{Change in Quantity Consumed}$

In simple form: $MU = dTU / dQ$

If quantity increases by one unit, then: $MU = TU \text{ of current unit} - TU \text{ of previous unit}$

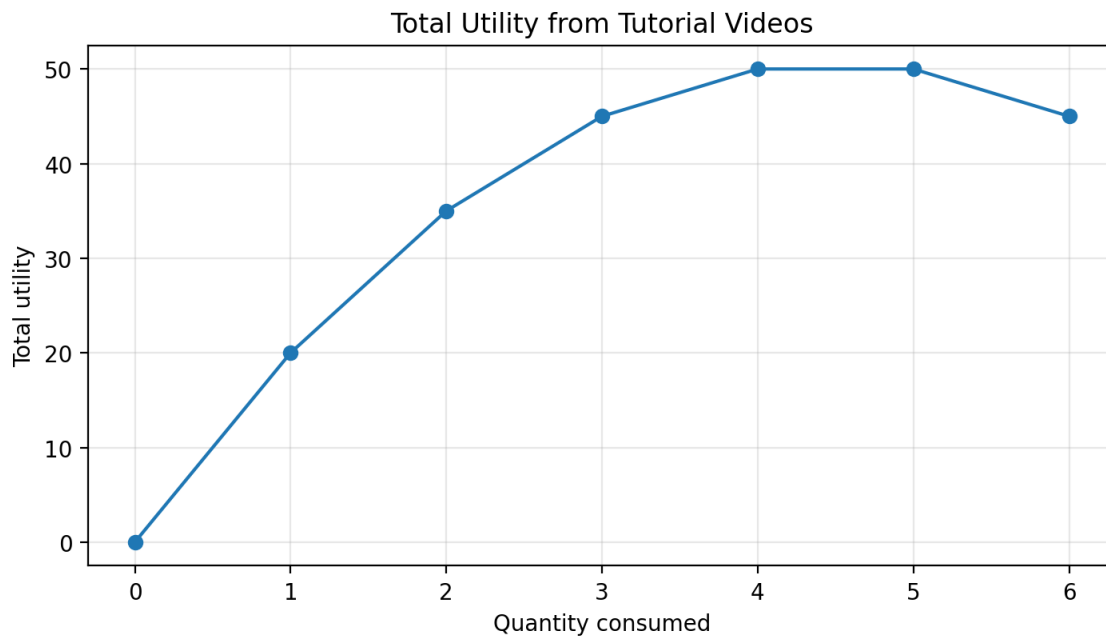
3. Example: Utility from Watching Programming Tutorial Videos

Suppose a CST student watches programming tutorial videos continuously.

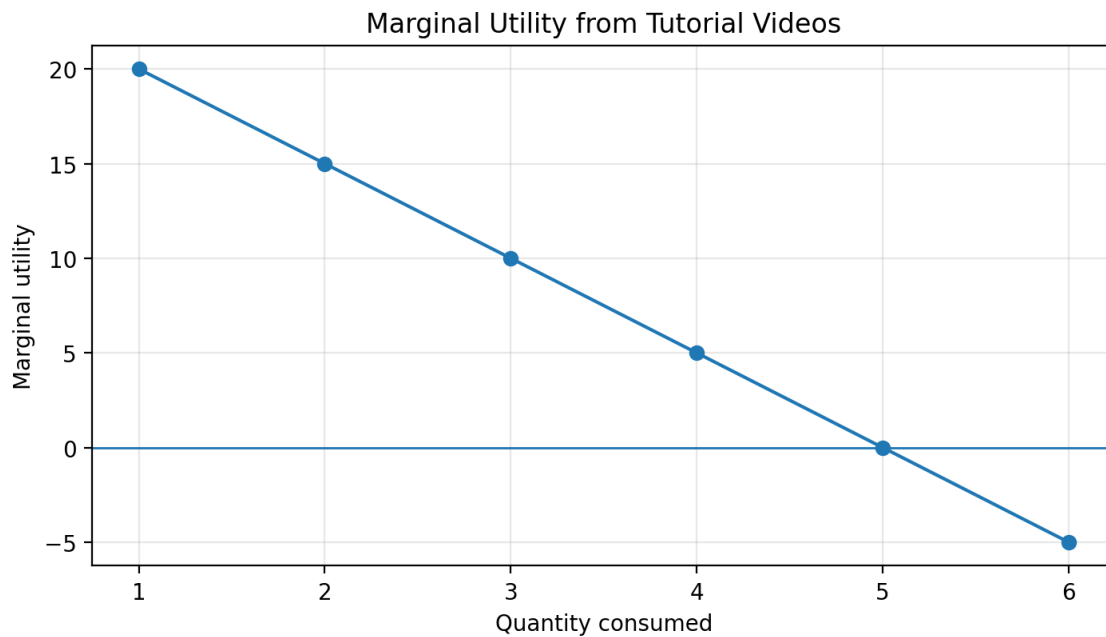
Number of Tutorial Videos Watched	Total Utility	Marginal Utility
0	0	-
1	20	20
2	35	15
3	45	10
4	50	5
5	50	0
6	45	-5

Explanation

The first video gives high satisfaction because the student learns something new. The second video also gives satisfaction, but less than the first. After watching many videos continuously, the student becomes tired. The fifth video adds no extra satisfaction. The sixth video may even reduce satisfaction because of fatigue or confusion.



Graph 1: Total utility first increases, then reaches a maximum, and may later fall.



Graph 2: Marginal utility declines as more units are consumed and may become zero or negative.

4. Law of Diminishing Marginal Utility

The law of diminishing marginal utility states that, other things remaining constant, as a consumer consumes more and more units of a good, the additional satisfaction from each extra unit gradually decreases.

In simple words: The more we consume of the same good at a time, the less extra satisfaction we get from each additional unit.

Example

If a thirsty student drinks one glass of water, the first glass gives very high satisfaction. The second glass gives satisfaction, but less than the first. The third gives even less. After several glasses, the student may not want more water.

5. Graph: Total Utility and Marginal Utility

The total utility graph and marginal utility graph should be explained together because marginal utility is the change in total utility. Total utility first rises at a decreasing rate, becomes maximum when marginal utility is zero, and then may fall when marginal utility becomes negative.

In the example of tutorial videos, total utility rises from 0 to 50 as the student watches the first five videos. It remains at 50 at the fifth video because marginal utility is zero. At the sixth video, total utility falls to 45 because marginal utility is negative.

Marginal utility falls from 20 to -5. This downward movement shows that each additional tutorial video provides less additional satisfaction than the previous one when watched continuously.

6. Relationship Between Total Utility and Marginal Utility

Situation	Total Utility	Marginal Utility
When TU is increasing	TU is rising	MU is positive: additional units add satisfaction
When TU is maximum	TU reaches its highest point	MU is zero: extra unit adds no satisfaction
When TU is decreasing	TU is falling	MU is negative: extra unit reduces satisfaction

Important Point

Total utility and marginal utility are closely related. Marginal utility explains the change in total utility.

Note

If a student watches tutorial videos but takes proper breaks, the decline in marginal utility may be slower. But if the student watches many videos without rest, marginal utility may fall quickly.

8. Importance of the Law of Diminishing Marginal Utility

This law is important because it helps explain:

1. Why demand curves usually slope downward.
2. Why consumers do not buy unlimited quantities of one good.
3. Why variety is important in consumption.
4. Why price must fall if sellers want consumers to buy more.
5. Why consumers allocate income among different goods.

Example for CST Students

A student may value the first 10 GB of mobile data highly. But after having enough data for classes, assignments, and communication, extra data may provide less additional satisfaction. Therefore, the student may not be willing to pay the same high price for every additional GB.

9. From Utility to Demand

Utility is closely connected to demand.

A consumer demands a good because the good gives utility. However, as marginal utility falls, the consumer becomes willing to pay less for additional units.

This helps explain the law of demand: When price falls, consumers usually buy more. When price rises, consumers usually buy less.

Example

If the price of a mobile data package falls, students may buy more data. If the price rises, they may reduce usage or switch to cheaper packages.

10. CST-Focused Applications of Diminishing Marginal Utility

10.1 Online Learning

The first hour of an online programming course may be very useful. The fourth or fifth continuous hour may be less useful because attention decreases.

10.2 Mobile Data

The first few GB of data may be very important for study and communication. Extra data used only for entertainment may provide less marginal utility.

10.3 Smartphone Features

A better camera, faster processor, or more RAM may initially add utility. But after a certain level, additional improvements may give smaller extra satisfaction to an average user.

10.4 Software Subscriptions

The first useful software subscription may provide high utility. But if a student subscribes to too many tools, the additional utility from each new subscription may decline.

Demand, Law of Demand, Demand Schedule and Demand Curve

1. Meaning of Demand

In economics, demand means the quantity of a good or service that consumers are willing and able to buy at a particular price during a particular period of time.

Demand has three elements:

1. Desire for the good.
2. Ability to pay for the good.
3. Willingness to pay for the good.

Without purchasing power, desire is not demand.

Example

A student may want a high-end gaming laptop. But if the student cannot afford it, this is only a desire, not effective demand.

If the student has enough money and is willing to buy it at the current price, then it becomes demand.

2. Demand Function

A demand function shows the relationship between demand and its determinants.

A simple demand function can be written as: $Q_d = f(P)$

Where: Q_d = quantity demanded; P = price of the good.

A more complete demand function is: $Q_d = f(P, Y, P_r, T, E, N)$

Symbol	Meaning
P	Price of the good
Y	Consumer income
P_r	Price of related goods
T	Taste and preference
E	Expectation about future price
N	Number of consumers

3. Law of Demand

The law of demand states that, other things remaining constant, when the price of a good falls, its quantity demanded rises; and when the price rises, its quantity demanded falls.

In simple words: Price and quantity demanded are inversely related.

Example

If the price of a mobile data package falls, students may buy more data. If the price rises, students may buy less data or switch to Wi-Fi.

4. Assumptions of the Law of Demand

The law of demand works when other factors remain constant. This condition is called *ceteris paribus*, meaning other things remaining the same.

5. Why Does the Demand Curve Slope Downward?

The demand curve slopes downward because of several reasons.

5.1 Diminishing Marginal Utility

As consumers buy more units of a good, marginal utility falls. Therefore, they are willing to pay less for additional units.

Example: A student may pay a high price for essential internet data but may not pay the same price for extra data used only for entertainment.

5.2 Income Effect

When price falls, the consumer's real income increases. The consumer can buy more with the same money.

Example: If the price of an online course falls from Tk. 3,000 to Tk. 1,500, more students can afford it.

5.3 Substitution Effect

When the price of one good falls, it becomes cheaper compared to its substitutes. Consumers may shift toward the cheaper good.

Example: If the subscription price of one cloud storage service falls, students may switch from another service.

5.4 New Consumers Enter the Market

At lower prices, consumers who previously could not afford the good may start buying it.

Example: If laptop prices fall, more students may buy laptops.

6. Demand Schedule

A demand schedule is a table that shows different quantities demanded at different prices.

Example: Demand for Mobile Data Packages

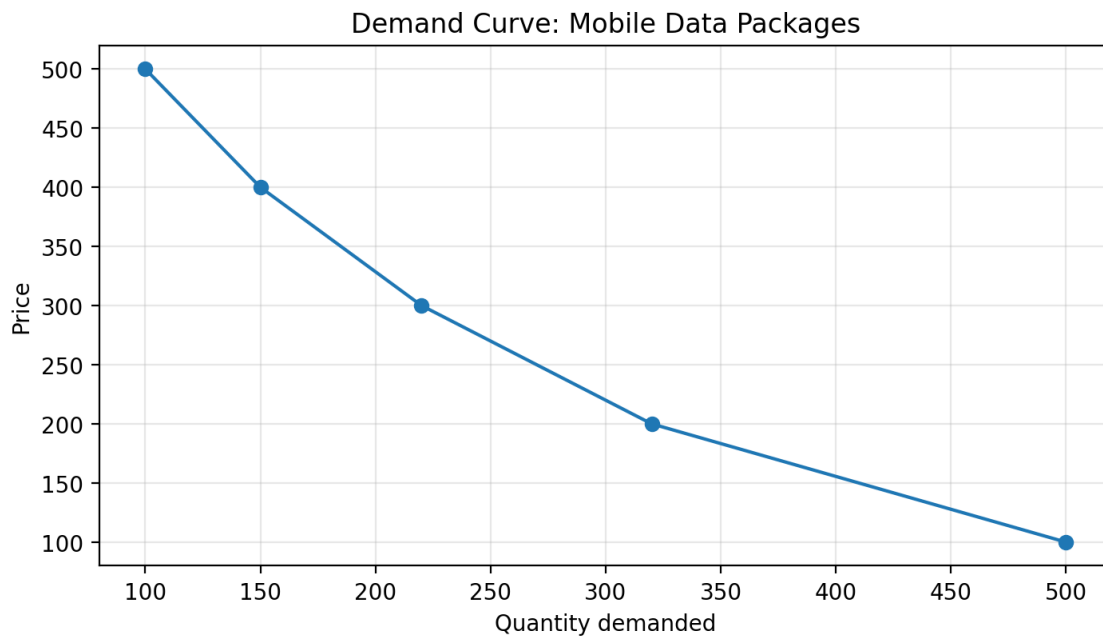
Price of Data Package	Quantity Demanded by Students
Tk. 500	100 packages
Tk. 400	150 packages
Tk. 300	220 packages
Tk. 200	320 packages
Tk. 100	500 packages

The table shows that when price falls, quantity demanded rises.

7. Demand Curve

A demand curve is a graphical representation of a demand schedule.

Price is usually measured on the vertical axis, and quantity demanded is measured on the horizontal axis.



Graph 3: Demand curve for mobile data packages. The demand curve slopes downward from left to right.

8. Linear Demand Example

Suppose the demand function for online courses is: $Q_d = 100 - 2P$

Where: Q_d = quantity demanded; P = price in hundred taka units.

If $P = 10$, then $Q_d = 100 - 2(10) = 80$.

If $P = 20$, then $Q_d = 100 - 2(20) = 60$.

If $P = 30$, then $Q_d = 100 - 2(30) = 40$.

Price	Quantity Demanded
10	80
20	60
30	40
40	20

This again shows an inverse relationship between price and quantity demanded.

9. Individual Demand and Market Demand

Individual Demand

Individual demand refers to the demand of one consumer.

Example: One student demands one online course at Tk. 1,000.

Market Demand

Market demand is the total demand of all consumers in the market.

Example: If 500 students each want one online course at Tk. 1,000, total market demand is 500 courses.

Price	Student A Demand	Student B Demand	Student C Demand	Market Demand
Tk. 500	1	2	1	4

Tk. 300	2	3	2	7
Tk. 100	3	5	4	12

10. Movement Along the Demand Curve

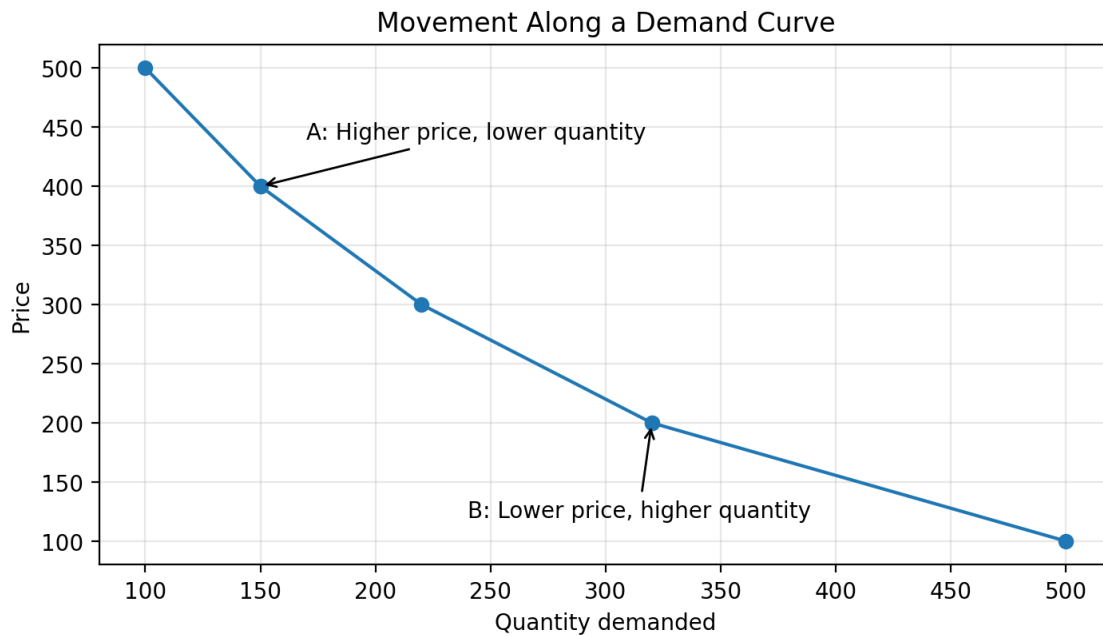
A movement along the demand curve happens only because of a change in the price of the good itself.

10.1 Expansion of Demand

When price falls and quantity demanded rises, it is called expansion of demand.

10.2 Contraction of Demand

When price rises and quantity demanded falls, it is called contraction of demand.



Graph 4: Movement along the demand curve. Moving from A to B shows expansion of demand because price falls and quantity demanded rises.

11. Shift of the Demand Curve

A shift of the demand curve happens when demand changes due to factors other than the price of the good itself.

Causes of Increase in Demand

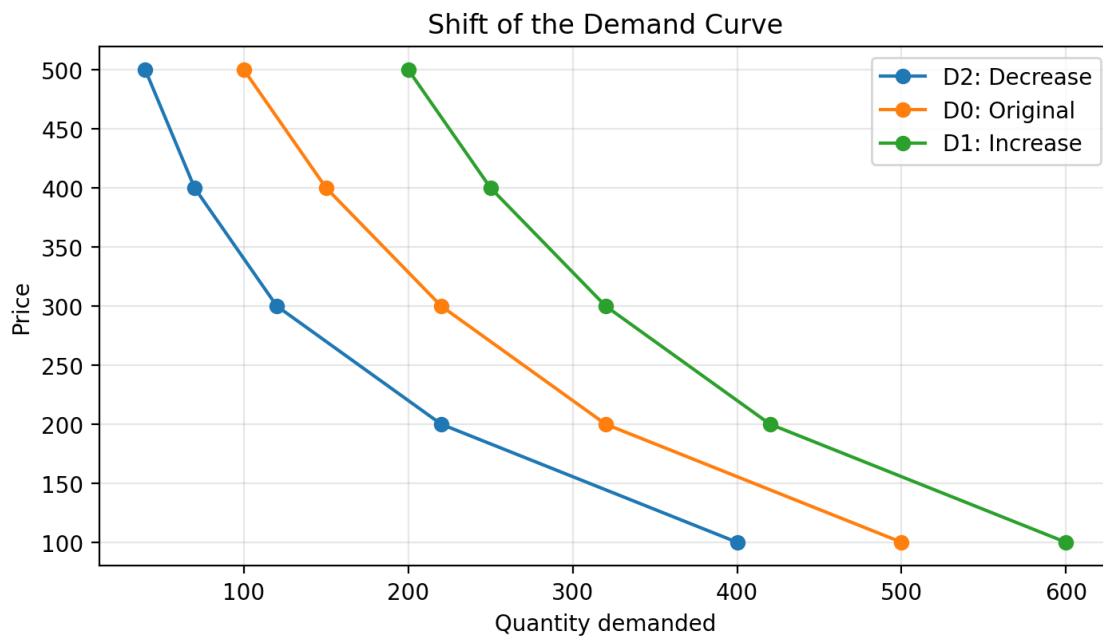
Demand may increase because of:

1. Increase in income.
2. Increase in preference for the good.
3. Increase in price of substitute goods.
4. Fall in price of complementary goods.
5. Expectation of future price rise.
6. Increase in number of consumers.

Causes of Decrease in Demand

Demand may decrease because of:

1. Fall in income.
2. Decrease in preference for the good.
3. Fall in price of substitute goods.
4. Rise in price of complementary goods.
5. Expectation of future price fall.
6. Decrease in number of consumers.



Graph 5: Shift of demand curve. Shift to the right indicates increase in demand; shift to the left indicates decrease in demand.

12. CST-Focused Demand Examples

12.1 Demand for Laptops

Demand for laptops may increase when:

- CST enrollment increases.
- Online classes become common.
- Programming courses require laptops.
- Laptop prices fall.
- Student income or family income increases.
- Freelancing opportunities increase.

Demand may decrease when:

- Prices rise.
- Alternative devices become cheaper.
- Students use university labs instead.

- Income falls.

12.2 Demand for Mobile Data

Demand for mobile data depends on:

- Price of data packages,
- Online class requirements,
- Availability of Wi-Fi,
- Income of students,
- Entertainment habits,
- App usage,
- Quality of network.

12.3 Demand for AI Tools

Demand for AI tools may increase when:

- Students find them useful for coding and learning.
- Subscription prices fall.
- Teachers allow responsible use.
- Job markets reward AI-related skills.
- Free versions become limited and paid versions become more valuable.

Supply, Law of Supply, Supply Schedule, Supply Curve and Market Equilibrium

Part A: Supply

1. Meaning of Supply

In economics, supply means the quantity of a good or service that sellers are willing and able to sell at a particular price during a particular period of time.

Supply has three elements:

1. Willingness to sell.
2. Ability to sell.
3. A specific price and time period.

Example

A software firm may be willing to supply 100 website development services per month at Tk. 20,000 per project. If the price rises to Tk. 30,000, the firm may be willing to supply more services.

2. Supply Function

A simple supply function can be written as: $Q_s = f(P)$

Where: Q_s = quantity supplied; P = price of the good.

A more complete supply function is: $Q_s = f(P, C, T, G, E, N)$

Symbol	Meaning
P	Price of the good
C	Cost of production
T	Technology
G	Government policy or tax/subsidy
E	Expectation about future price
N	Number of sellers

3. Law of Supply

The law of supply states that, other things remaining constant, when the price of a good rises, quantity supplied rises; and when the price falls, quantity supplied falls.

In simple words: Price and quantity supplied are positively related.

Example

If the payment for website development increases, more freelancers may be willing to offer website development services.

The law of supply works when other factors remain constant.

5. Why Does the Supply Curve Slope Upward?

The supply curve usually slopes upward because:

5.1 Higher Price Means Higher Profit

When price rises, producers can earn more profit, so they supply more.

5.2 Higher Price Covers Higher Cost

To produce more, firms may need to use more resources, hire more workers, or pay overtime. Higher price helps cover these costs.

5.3 New Sellers Enter the Market

At higher prices, new sellers may enter the market.

Example: If app development becomes highly profitable, more developers may enter the app market.

6. Supply Schedule

A supply schedule is a table showing different quantities supplied at different prices.

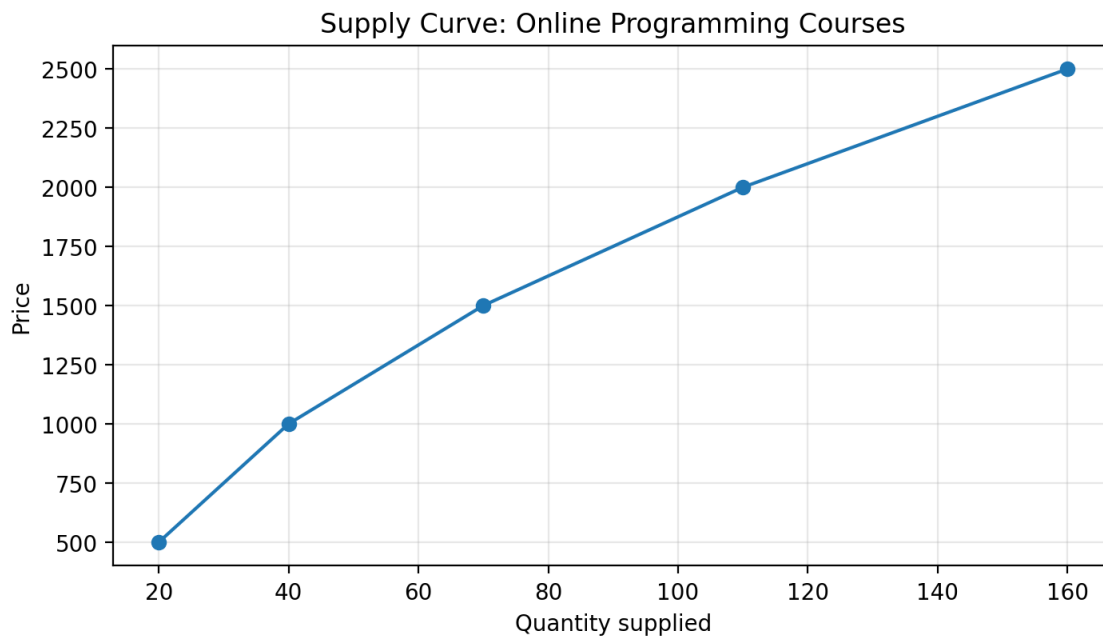
Example: Supply of Online Programming Courses

Price per Course	Quantity Supplied
Tk. 500	20 courses
Tk. 1,000	40 courses
Tk. 1,500	70 courses
Tk. 2,000	110 courses
Tk. 2,500	160 courses

The table shows that when price rises, quantity supplied rises.

7. Supply Curve

A supply curve is a graphical representation of a supply schedule.



Graph 6: Supply curve for online programming courses. The supply curve slopes upward from left to right.

8. Movement Along the Supply Curve

Movement along the supply curve happens because of a change in the price of the good itself.

Extension of Supply

When price rises and quantity supplied rises, it is called extension of supply.

Contraction of Supply

When price falls and quantity supplied falls, it is called contraction of supply.

9. Shift of Supply Curve

A shift in supply happens because of factors other than the price of the good itself.

Causes of Increase in Supply

Supply may increase because of:

1. Improvement in technology.
2. Fall in cost of production.
3. Increase in number of sellers.
4. Government subsidy.
5. Better infrastructure.
6. Easier access to inputs.

Causes of Decrease in Supply

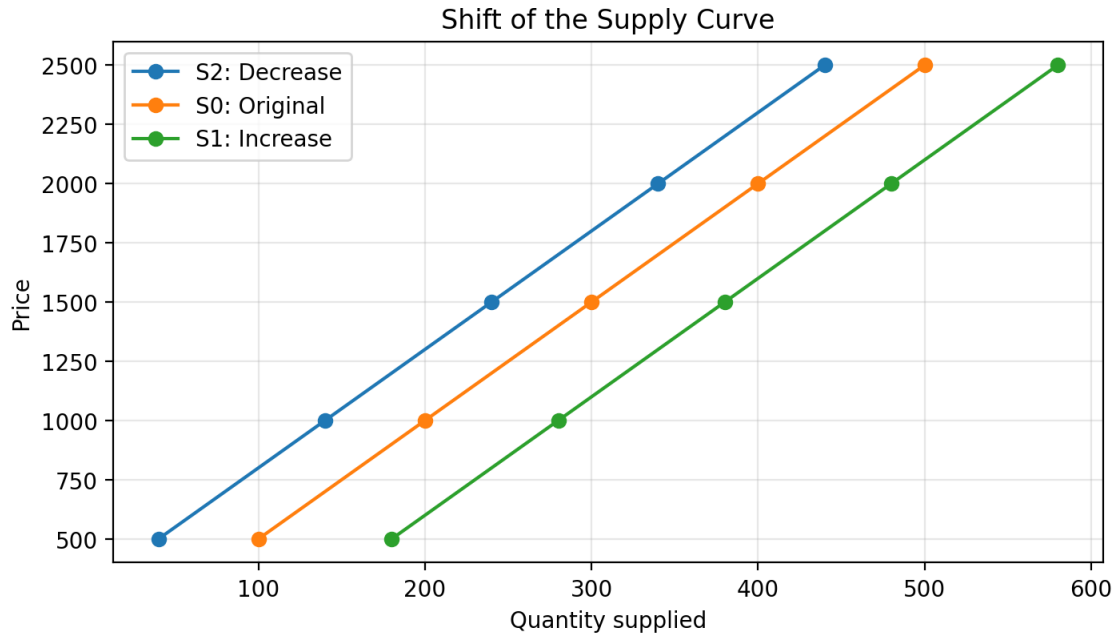
Supply may decrease because of:

1. Rise in production cost.
2. Higher tax.

3. Fall in number of sellers.
4. Technical problems.
5. Natural disaster.
6. Shortage of inputs.

CST Example

If cloud server costs fall, app developers can supply more digital services at the same price. This shifts the supply curve to the right.



Graph 7: Shift of supply curve. Shift to the right indicates increase in supply; shift to the left indicates decrease in supply.

Part B: Market Equilibrium

10. Meaning of Market Equilibrium

Market equilibrium occurs when quantity demanded equals quantity supplied at a particular price.

At equilibrium: Quantity Demanded = Quantity Supplied

The equilibrium price is the price at which buyers are willing to buy exactly the amount sellers are willing to sell.

The equilibrium quantity is the quantity bought and sold at the equilibrium price.

11. Demand and Supply Schedule Together

Example: Market for Online Programming Courses

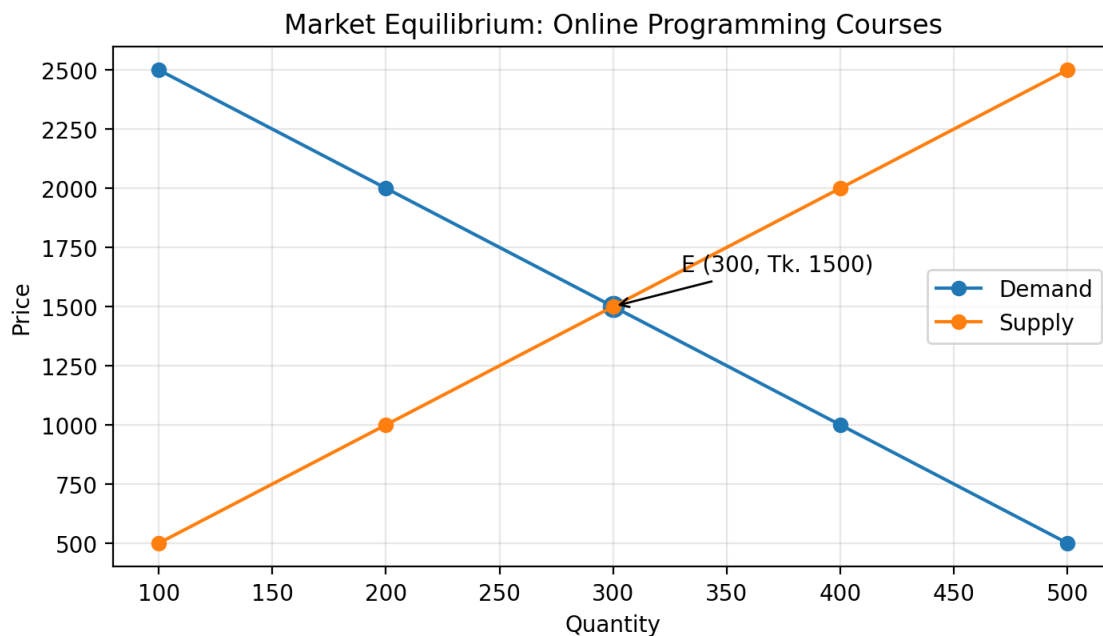
Price	Quantity Demanded	Quantity Supplied	Market Situation
Tk. 500	500	100	Shortage
Tk. 1,000	400	200	Shortage
Tk. 1,500	300	300	Equilibrium
Tk. 2,000	200	400	Surplus

Tk. 2,500	100	500	Surplus
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At Tk. 1,500: Quantity demanded = 300 and Quantity supplied = 300.

Therefore, the equilibrium price is Tk. 1,500, and the equilibrium quantity is 300 courses.

12. Graph: Market Equilibrium



Graph 8: Market equilibrium occurs where the demand curve and supply curve intersect.

13. Shortage and Surplus

Shortage

A shortage occurs when quantity demanded is greater than quantity supplied.

$$Q_d > Q_s$$

This usually happens when price is below equilibrium.

Example: If an online course is very cheap, many students want to enroll, but instructors may not supply enough seats.

Surplus

A surplus occurs when quantity supplied is greater than quantity demanded.

$$Q_s > Q_d$$

This usually happens when price is above equilibrium.

Example: If a course is too expensive, many instructors may offer courses, but fewer students will buy them.

14. How Price Adjusts in the Market

When There Is Shortage

If there is a shortage, buyers compete for limited goods. This puts upward pressure on price. As price rises, quantity demanded falls and quantity supplied rises until equilibrium is reached.

When There Is Surplus

If there is a surplus, sellers compete to attract buyers. This puts downward pressure on price. As price falls, quantity demanded rises and quantity supplied falls until equilibrium is reached.

15. Numerical Example of Market Equilibrium

Suppose the demand and supply functions for an online programming course are:

$$Q_d = 100 - 2P$$

$$Q_s = 20 + 2P$$

At equilibrium: $Q_d = Q_s$

$$\text{So, } 100 - 2P = 20 + 2P$$

$$100 - 20 = 2P + 2P$$

$$80 = 4P$$

$$P = 20$$

Now substitute $P = 20$ into either demand or supply equation:

$$Q_d = 100 - 2(20) = 60$$

$$Q_s = 20 + 2(20) = 60$$

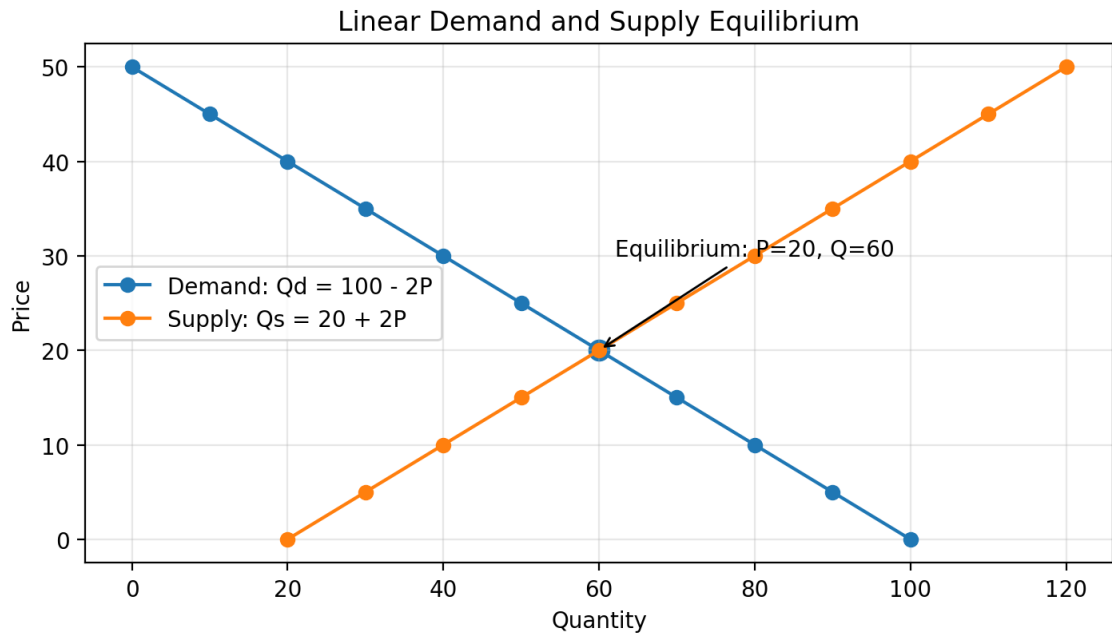
Therefore: Equilibrium Price = 20 and Equilibrium Quantity = 60.

Here, if price is measured in hundred taka, then equilibrium price is: $20 \times 100 = \text{Tk. } 2,000$.

16. Graph of Linear Demand and Supply

For the above example: Demand: $Q_d = 100 - 2P$; Supply: $Q_s = 20 + 2P$.

The equilibrium occurs at: $P = 20$, $Q = 60$.



Graph 9: Linear demand and supply equilibrium for $Q_d = 100 - 2P$ and $Q_s = 20 + 2P$.

17. Changes in Market Equilibrium

Equilibrium changes when demand or supply shifts.

17.1 Increase in Demand

If demand increases and supply remains the same:

- Equilibrium price rises.
- Equilibrium quantity rises.

Example: If more CST students start learning programming, demand for programming courses increases. Course price and enrollment may rise.

17.2 Decrease in Demand

If demand decreases and supply remains the same:

- Equilibrium price falls.
- Equilibrium quantity falls.

Example: If students shift from paid courses to free YouTube tutorials, demand for paid courses may fall.

17.3 Increase in Supply

If supply increases and demand remains the same:

- Equilibrium price falls.
- Equilibrium quantity rises.

Example: If many instructors start offering online programming courses, supply increases. Course prices may fall and more students may enroll.

17.4 Decrease in Supply

If supply decreases and demand remains the same:

- Equilibrium price rises.
- Equilibrium quantity falls.

Example: If server costs rise or skilled instructors become unavailable, supply of quality online courses may decrease.

CST-Focused Applications

1. Market for Laptops

Demand increases when:

- More students enroll in CST.
- Online learning increases.
- Programming courses require personal laptops.
- Freelancing opportunities increase.

Supply increases when:

- Import costs fall.
- Technology improves.

- More brands enter the market.
- Distribution channels improve.

Equilibrium price depends on both demand and supply.

2. Market for Mobile Data

Demand side:

- Online classes,
- Social media,
- Streaming,
- App use,
- Freelancing,
- Remote work.

Supply side:

- Telecom infrastructure,
- Spectrum availability,
- Competition among operators,
- Network technology,
- Government regulation.

If demand rises faster than supply, data prices may rise or network congestion may occur.

3. Market for Software Developers

Demand for software developers increases when:

- More firms need websites and apps.
- Digital transformation expands.
- Banks, NGOs, and government services go online.
- AI and data analytics become important.

Supply of software developers increases when:

- More students learn programming.
- Training centers expand.
- Online learning becomes cheaper.
- Universities improve CST education.

Equilibrium wage depends on the demand and supply of skills.

4. Market for Cloud Storage

Demand increases when users store more photos, videos, documents, and backups.

Supply increases when cloud providers invest in servers, storage, data centers, and network infrastructure.

Cloud storage is a useful CST example because it seems digital, but it depends on physical infrastructure such as servers, electricity, cooling, and skilled engineers.

Graph 1: Total Utility and Marginal Utility

Use tutorial videos, cups of tea, or mobile data as the example.

Quantity	Total Utility	Marginal Utility
1	20	20
2	35	15
3	45	10
4	50	5
5	50	0
6	45	-5

Explain:

- TU rises when MU is positive.
- TU is maximum when MU is zero.
- TU falls when MU is negative.

Graph 2: Demand Curve

Price	Quantity Demanded
500	100
400	150
300	220
200	320
100	500

Explain:

- Price on vertical axis.
- Quantity on horizontal axis.
- Demand curve slopes downward.

Graph 3: Supply Curve

Price	Quantity Supplied
500	20
1,000	40
1,500	70
2,000	110
2,500	160

Explain:

- Price on vertical axis.
- Quantity on horizontal axis.
- Supply curve slopes upward.

Graph 4: Market Equilibrium

Price	Quantity Demanded	Quantity Supplied	Situation
500	500	100	Shortage
1,000	400	200	Shortage
1,500	300	300	Equilibrium
2,000	200	400	Surplus
2,500	100	500	Surplus

Explain:

- Equilibrium occurs where $Q_d = Q_s$.
- Below equilibrium price, shortage occurs.
- Above equilibrium price, surplus occurs.